TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology
Wardha Road, Nagpur - 441108
Accredited with NAAC A+ Grade
Approved by AICTE, New Delhi, Govt. of Maharashtra

(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)
Department of Information Technology
Session 2023-24(Even Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

END SEMESTER
Semester: IV
Programme: B. Tech Information Technology
Course: Database Management System

Time: 3 Hours **Max Marks:** 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT32402	Course Outcome
CO1	Understand the fundamental of database system and DBMS concepts.
CO2	Identify the concept of conceptual data modeling using the E/R model.
CO3	Illustrate the concept of embedded SQL and its role in application development.
CO4	Analyze the process of database normalization and the steps for achieving 1NF,2NF,3NF,and BCNF
CO5	Compare Various indexing techniques like hash-based indexing,dynamic hashing,and B+trees.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Define DBMS.	CO1	2	2
b.	What is an entity?	CO2	2	2
c.	What is SQL?	CO3	2	2
d.	Define normalization.	CO4	3	2
e.	What is a transaction?	CO5	2	2
Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks

a.	Explain three-schema architecture of DBMS.	CO1	2	5
b.	Describe DBMS components in detail.	CO1	2	5
c.	Explain advantages of DBMS over file system.	C01	2	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain E-R model with diagram.	CO2	2	5
b.	Describe relational algebra operations.	CO2	3	5
c.	Explain conversion of E-R model to relational schema.	CO2	3	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain SQL data definition and manipulation commands.	CO3	2	5
b.	Describe nested queries with examples.	CO3	3	5
c.	Explain GROUP BY and HAVING clause.	CO3	4	5
Que 5.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain functional dependencies and types.	CO4	2	5
b.	Describe normalization up to 3NF.	CO4	3	5
c.	Explain BCNF with example.	CO4	4	5
Que 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain indexing techniques (B+ tree).	CO5	2	5
b.	Describe ACID properties.	CO5	3	5
c.	Explain concurrency control using locking.	CO5	4	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create